Broader conductor gaps: sizes, precision and remaining work

2026-10-26. **Parameter search and experimental design; zero new ECDLP solves and zero new explicit curve models.** Native source was frozen before each search. The corrected crater result remains inconclusive: S2/S1 = 1.01112520, enumerated 98% interval [0.97323,1.05706].

The useful next objective is a larger measured range with enough independent replication to exclude an average rho cost difference beyond ±5%. Merely increasing the conductor integer, subgroup bits or repeated seeds cannot guarantee that conclusion. The user requested broader gaps and stronger confidence; the completed parameter work and pending measurements are distinguished below.

Requirement	Actual status
Broaden candidate conductor gaps and subgroup sizes	Verified parameter arithmetic/primality: 128 native-size classes plus six upper classes.
Include much larger key-space parameters	Verified prime subgroup-order parameters through 254 bits; no supplied curve/generator or solve at that size.
Define a high-confidence comparison	Proposed: all four rung intervals within ±5%, nominal 99% simultaneous coverage; pilot calibration still required.
Construct exact models, floors and transported targets	Not performed; explicit map/order/stratum certificates remain required.
Run the disjoint pilot and confirmation	Not performed; native driver, replay, variance and resource feasibility remain blocked.
Establish broader equivalence or hardness	Not established. The earlier measured crater verdict is unchanged.

The retained parameter range

Each native-size rung has eight distinct classes for each of D=-11,-19, -43,-67: **32 classes per rung**, 128 total. Cofactor is at most 64; the actual subgroup bits, rather than field bits, control generic work. The design keeps depth four and the within-class S1/three-S2 comparison. Different classes hold different fields/traces; each eventual within-class comparison must hold field, order and subgroup fixed.

Field bits	Classes	Actual subgroup bits	Prime gap range
40	32	34–38	15,749–39,323
47	32	41–45	179,957–441,319
54	32	48–52	2,044,391–5,042,963
61	32	55–59	23,177,047–57,063,697
112	2	110	approximately 1.1004e15–1.3735e15
160	2	158	approximately 1.8462e22–2.3045e22
256	2	254	approximately 5.1965e36–6.4866e36

The upper classes use D=-43,-67, trace -2 and cofactor four. Exact decimal integers and witnesses are in candidates.csv; design.json records all ranges and search counts. The independent verifier receipt covers all 134 classes and 22 cells. It explicitly verifies **zero models, maps or solver records**. Prime subgroup-order parameters do not become registered ICV1 curve identities until exact models and generators are constructed.

The old replication used gaps 997–4,799 and 25–32-bit subgroups. The original transported depth-six class had gap 631,409 and a 41-bit subgroup. Separate historical 53-bit ECDLP comparisons already cover gaps 33,555,419–35,282,963, with their recorded transfer/timing limits. The older 96-bit construction certifies a floor at gap 125,000,000,302,579 but does not certify full target transport. These are different evidence types; the new parameter search cannot be called a new comparison at those sizes.

For an actual endomorphism conductor ratio ℓ dividing the Frobenius conductor, $\ell \le 2\sqrt{p}/\sqrt{|D_K|}$; an additional depth-four factor gives $\ell \le 2\sqrt{p}/(16\sqrt{|D_K|})$. Thus gap and field size cannot be varied independently over arbitrary ranges. The design requires paired comparisons within classes and reports every subgroup/ cofactor; it does not claim a field-independent causal gap effect.

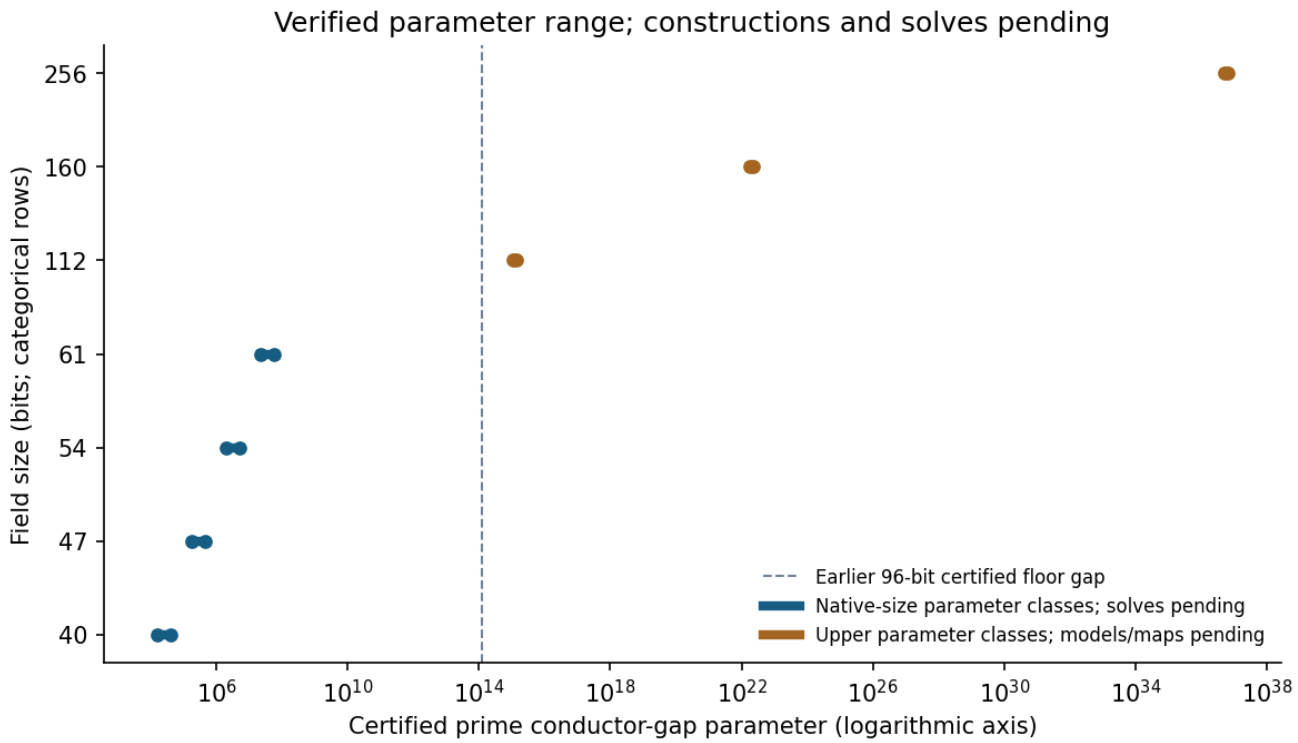


Figure 1: Verified parameter range. These marks are parameter certificates, not completed solver measurements.

Repetitions: a precision objective, not a magic count

The proposed primary comparison has four size rungs. Use 99.75% marginal intervals, giving at least 99% simultaneous coverage by the union bound **if their marginal coverage is valid**. Bootstrap intervals have approximate coverage and must be calibrated under the selected design. All four intervals must lie wholly inside [0.95,1.05]; failing to detect a difference is not evidence of equivalence. Individual classes and CM families also remain visible. A pooled average does not establish uniform equivalence.

The provisional allocation is 32 classes/rung, four craters/class, 1,000 distinct fresh logarithms/crater, two independent rho seeds per log, four rungs: **1,024,000 independent one-target solves**, plus full separate replays and controls. There is no shared multi-target collision table. Classes, target draws, solver seeds and correctness replays are distinct units. Replaying an existing record adds no statistical observation.

Native planning uses the old variance components, with variance of the log-ratio approximated by $\tau^2/K + (V_{\text{target}} + V_{\text{seed}}/R)/(KM)$. The assumed true ratio below is the old 1.01112520. The joint-power lower bound uses the union bound and does not assume independence across rungs.

Variance relative to old fitted model	Approximate per-rung power	Joint-power lower bound
1×	99.67%	98.68%
2×	84.97%	39.90%
4×	39.65%	0%

At ratio 1.01 the old-model joint lower bound is 99.22%; none of these values is measured power at the larger sizes. A zero union lower bound does not mean zero actual joint power. The old model itself used only 32 two-seed targets per class, with six of eight estimated target covariances negative before truncation; its uncertainty is unquantified.

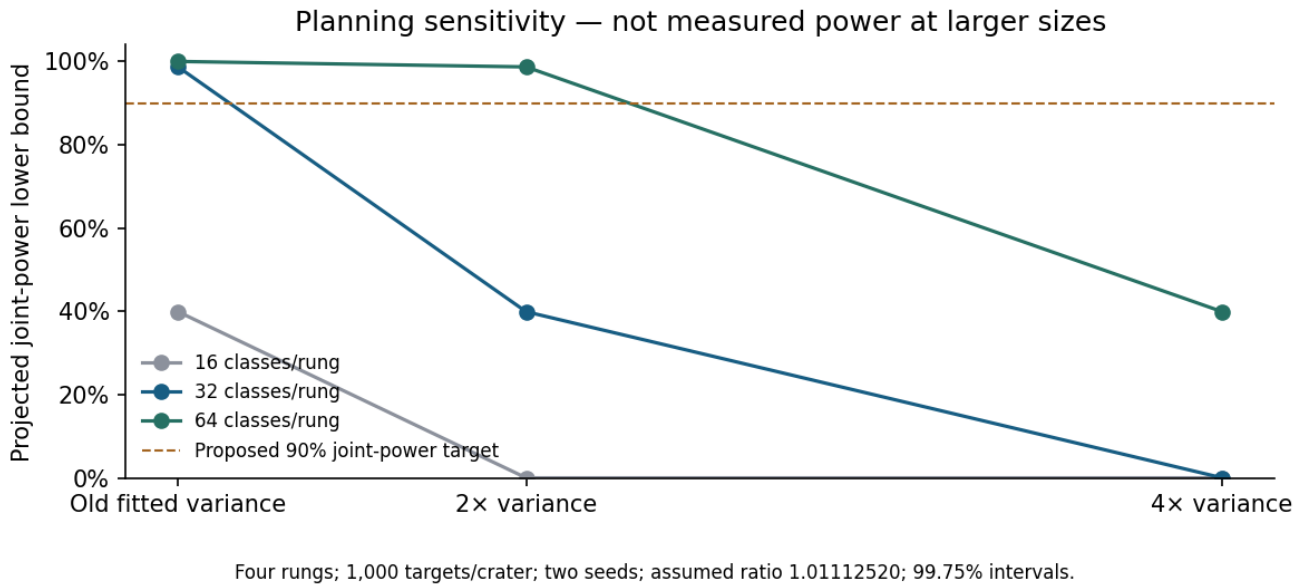


Figure 2: Conditional planning sensitivity. These projections use old small-group variance and are not validation of the larger campaign.

After model/map construction and execution-feasibility checks, run a separate pilot of 32 fresh targets/crater, two seeds, all 128 classes: **32,768 pilot solves**, plus separate replays. Exclude them from confirmation. The pilot must support at least 90% joint power for true ratios in [0.99,1.01], coverage calibration and the declared resource envelope before the confirmation count is frozen. Otherwise revise a new protocol and leave confirmation pending. Never continue until equivalence happens to pass, change the margin after results, or drop failed targets.

The native generic-negation model projects about 1.0051e14 group additions for the proposed native-size confirmation and 2.0102e14 including replays, before controls, construction and unpriced work. These are **operation-model projections, not measured runtime, complete costs or a resource allocation**. They make execution feasibility a real open requirement. The 254-bit subgroup parameters would imply roughly 2^{127} generic-rho work per solve; upper structural checks cannot stand in for completed rho comparisons.

Preserved failure and corrected family

The v1 card stopped after retaining 131 parameter classes: all 128 native classes and three upper D=-43 classes. All three D=-19 upper cells exhausted 3,000,000 candidates. No independent verification command ran after that stop. Raw v1 results, logs, receipt and stopped assessment remain unchanged.

The restricted D=-19, trace=-2, h=4 formula gives $p = 1216\ell^2 + 1$ and $r = 304\ell^2 + 1$. For prime $\ell > 5$, ell modulo 5 is 1,2,3 or 4; its square is 1 or 4. Respectively r or p is a proper multiple of 5. This is a four-case modular obstruction for this formula, not a statement about all D=-19 curves. The native residue table records each case. Supporting product-decomposition certificate **IDC1h4e934fe294c7d1ba** is retained, with 32 issued and 64 additional checked points over F_(2^61-1). That finite-field evaluation is an additional arithmetic check; the integer modular argument is the four explicit cases. Lean/lake are unavailable; no checked Lean proof is claimed.

The v2 amendment, frozen before execution, replaced only those three cells with D=-67. It reused every original row byte-for-byte, searched three new upper cells, then independently verified all 134 classes. The v2 receipt pins the source commit 834cc5015d221c9bac774bd9eb278db624db4d85, inputs, binaries, commands and outputs. The director's assessments concern execution and verbatim relay only, not the scientific conclusion.

Validation, reproducibility and pending work

Native C++17/GMP search and independent verification use different Miller–Rabin base sets and modular-exponentiation implementations. Upper primality uses recursive Pocklington witnesses with sufficiently large known factors; probable-prime tests only filter candidates. Bad known factors, duplicated classes and truncated cohorts are rejected. Reversed row order gives the same verifier receipt; the retained v1 prefix matches byte-for-byte. Source/inputs and artifact hashes accompany the study.

Rust library validation: 4,494 passed, zero failed, 119 ignored. The new native identity example has two passing tests. All 49 site tests pass; all 49 validation-gate tests and the gate pass **after local hydration**. The parent artifact-store migration removes 15 raw inputs needed by two existing validation rows; the unhydrated gate fails. Local copies were restored from pre-migration Git with all 53 referenced digests verified, without recomputing or changing frozen bytes. A clean CI checkout still needs hydration; see validation context.

Rebuild the two native programs using the amendment's commands; then run the verifier against results-v2/candidates.csv into a fresh receipt. The mechanical cards preserve the complete search commands. Re-render:

```
study=research/conductor_gap_size_ladder_20261026
MPLCONFIGDIR=/tmp/conductor-gap-size-mpl python3 "$study/render.py"
```

Python only lays out recorded native values. No performance result, IC ranking, benchmark row or measured volcano edge changed. Historical graphs in the old studies, scoreboard, progress timeline and leaderboard retain their frozen findings; the new parameter-range and sensitivity figures belong here. The companion PDF is inspected separately.

The problem record and blocked backlog card retain the exact next obligations: actual models/maps, native resumable driver, independent replay, disjoint pilot, calibrated precision and an executable resource envelope. No solver campaign or paid resource has been launched. Complete CM setup accounting and a prime-field algebraic oracle remain open. Rho comparisons alone cannot establish a general ECDLP hardness gap or full-cost speed equivalence.